

AE462: Aircraft Design-II

End-Sem Presentation

Abaqus Analysis of Wingbox Structure
and Buckling Margin Evaluation

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Objective

1.To create a wingbox model using Abaqus.

2.To apply different loads:

- Engine load
- Pressure load
- Lift load

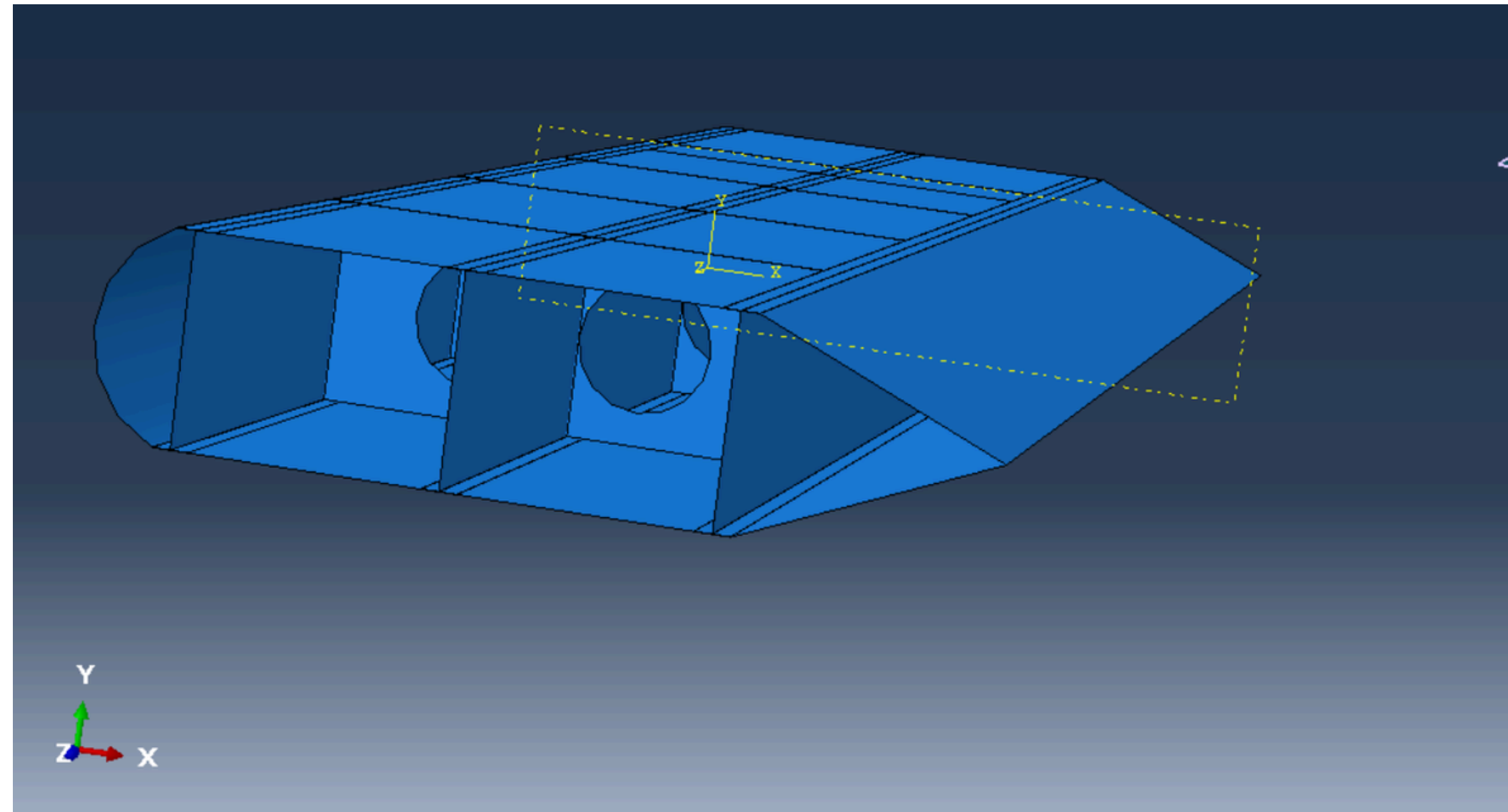
3.To study structural behavior:

- Stress distribution
- Displacement

4.To check if the structure is safe.

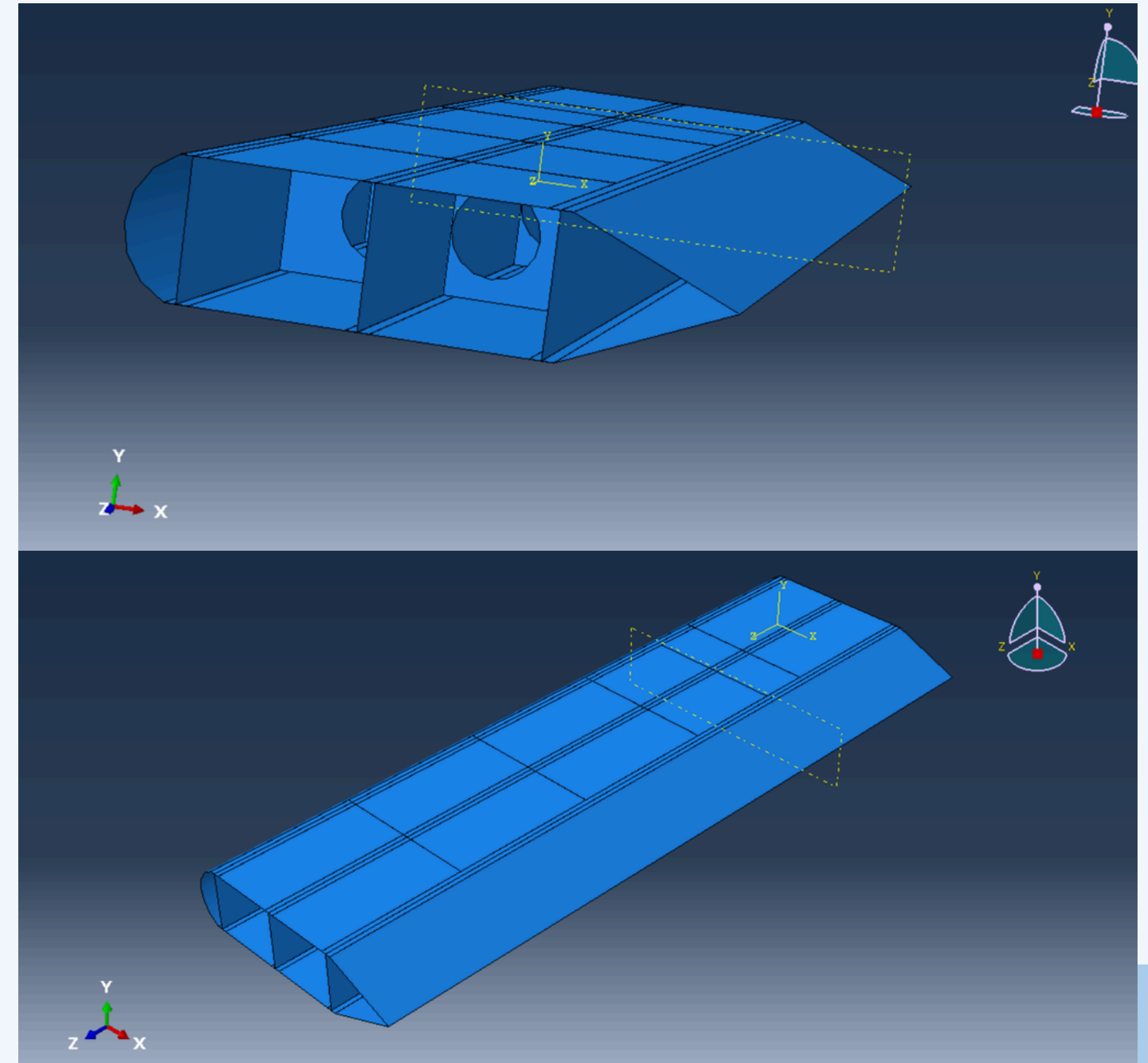
5.To calculate buckling for:

- Skin
- Spar web
- Spar flange
- Crippling



GEOMETRY OF WINGBOX

- Wingbox span= **17.03 m**
- Width = **0.457 m**
- Height (depth) = **0.12 m**
- Structure consists of:
 - Upper and lower skin
 - Spars (front and rear)
 - Ribs
- Rib spacing $\approx$ **1.5 m**



MATERIAL PROPERTIES

Aluminium:

Aluminium (Outer Structure)

- Young's Modulus, $E = 70 \text{ GPa}$
- Poisson's Ratio, $\nu = 0.3$
- Density, $\rho = 2700 \text{ kg/m}^3$
- Yield Strength, $\sigma_y = 310 \text{ MPa}$

Carbon Epoxy:

Carbon Epoxy (Composite)

- $E_1 = 162 \text{ GPa}$
- $E_2 = 7.93 \text{ GPa}$
- $G_{12} = 5.3 \text{ GPa}$
- $G_{13} = 5.3 \text{ GPa}$
- $G_{23} = 4 \text{ GPa}$
- $\nu_{12} = 0.35$
- $\rho = 1600 \text{ kg/m}^3$

Material properties defined in Abaqus model.

Composite Layup Definition

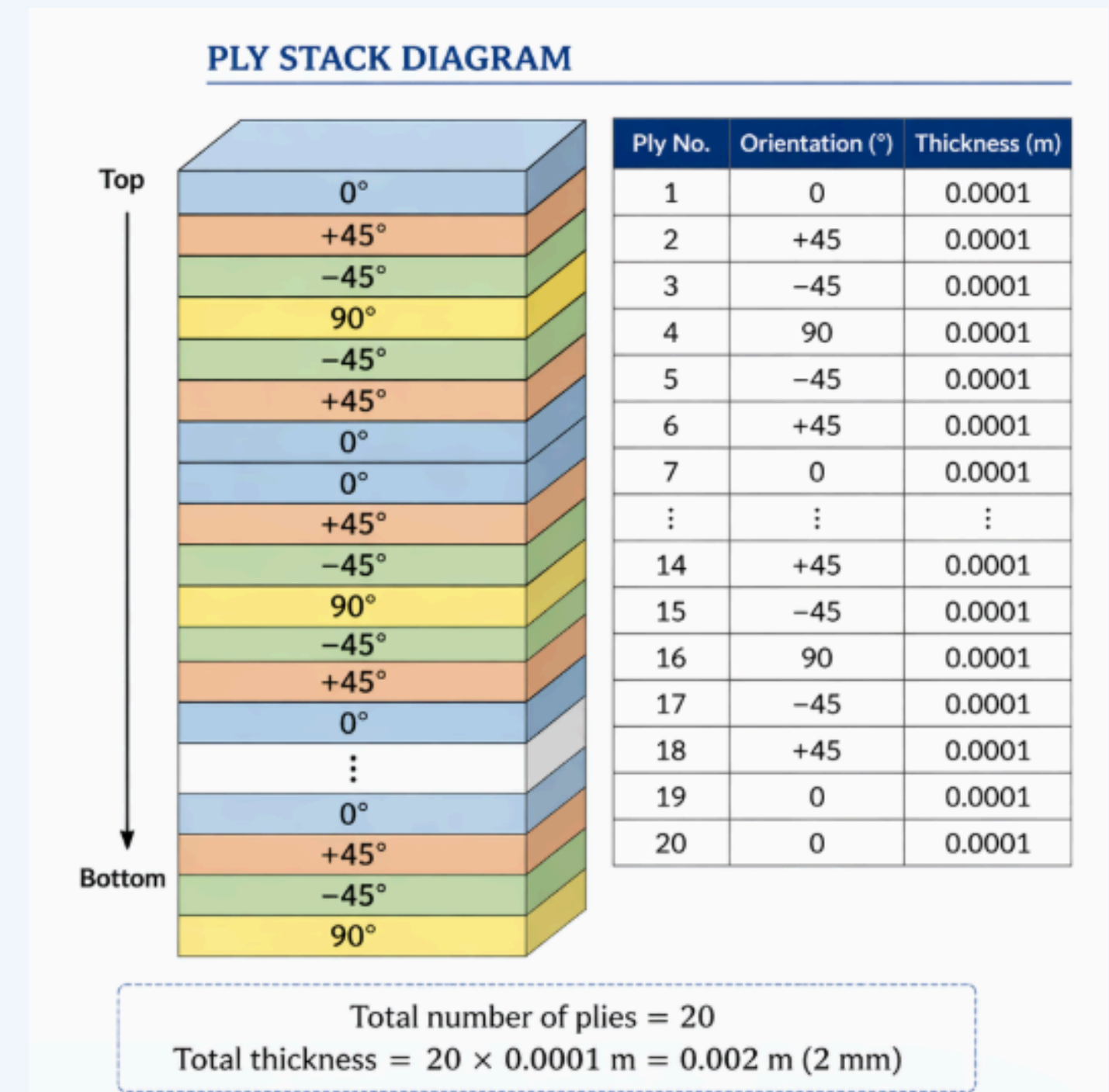
Main Content:

- Element type: Conventional shell (S4R, S3)
- Total number of plies: 20
- Ply thickness: 0.0001 m
- Total laminate thickness: 0.002 m

Orientation sequence:

- Stacking sequence includes:
 - 0°
 - $+45^\circ$
 - -45°
 - 90°

Symmetric laminate with balanced orientations.



Mesh and Element Details

Element Type:

- Shell elements used:
 - S4R (4-node quadrilateral, reduced integration)
 - S3 (3-node triangular)

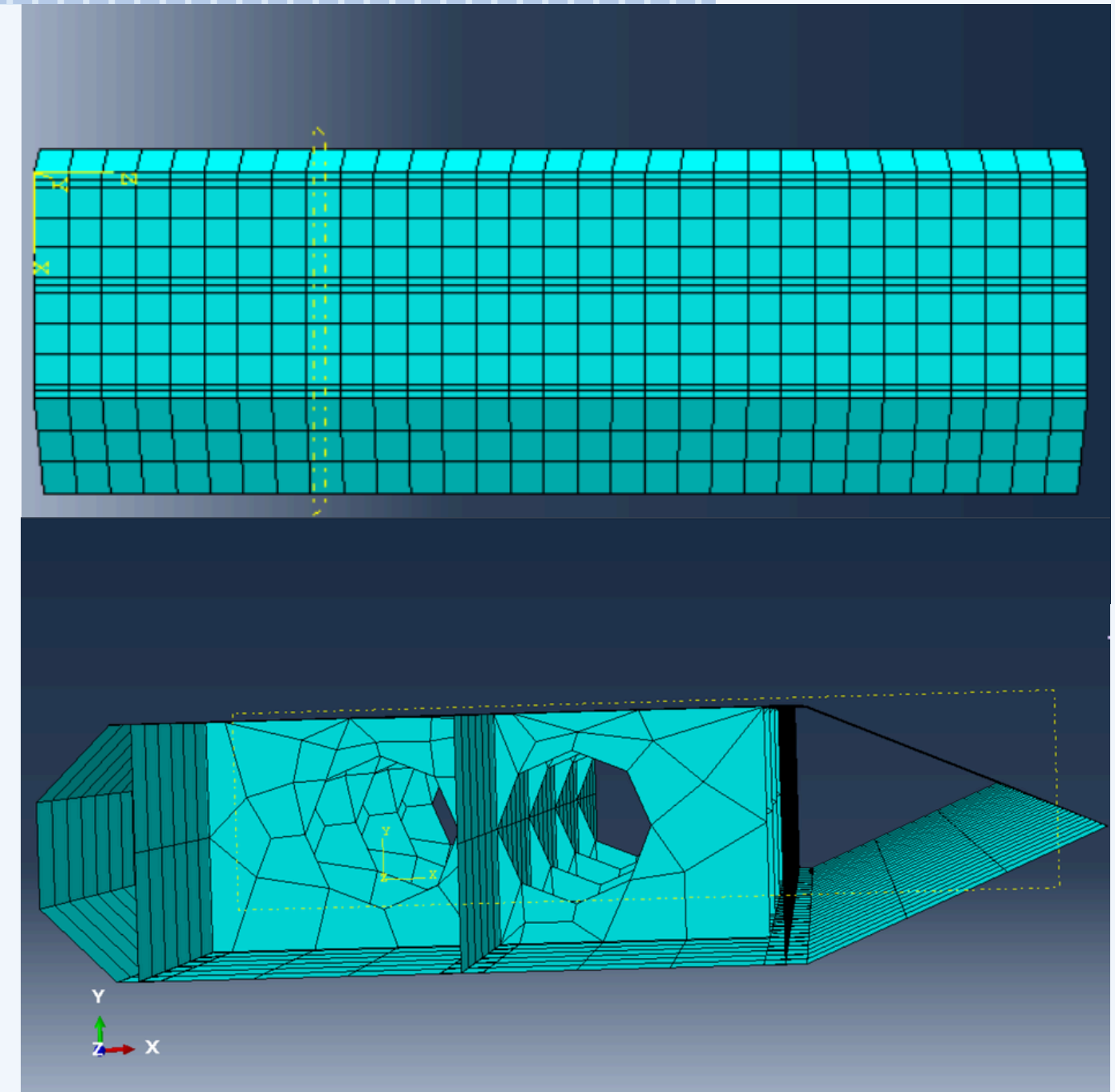
Shell elements are suitable for thin-walled structures like wingbox.

Mesh Details:

- Global mesh size: 0.48
- Total elements: ~1093
- Total nodes: ~980

Mesh Quality:

- Structured mesh used in most regions
- Triangular elements used in complex areas
- Mesh refined near ribs and geometry transitions



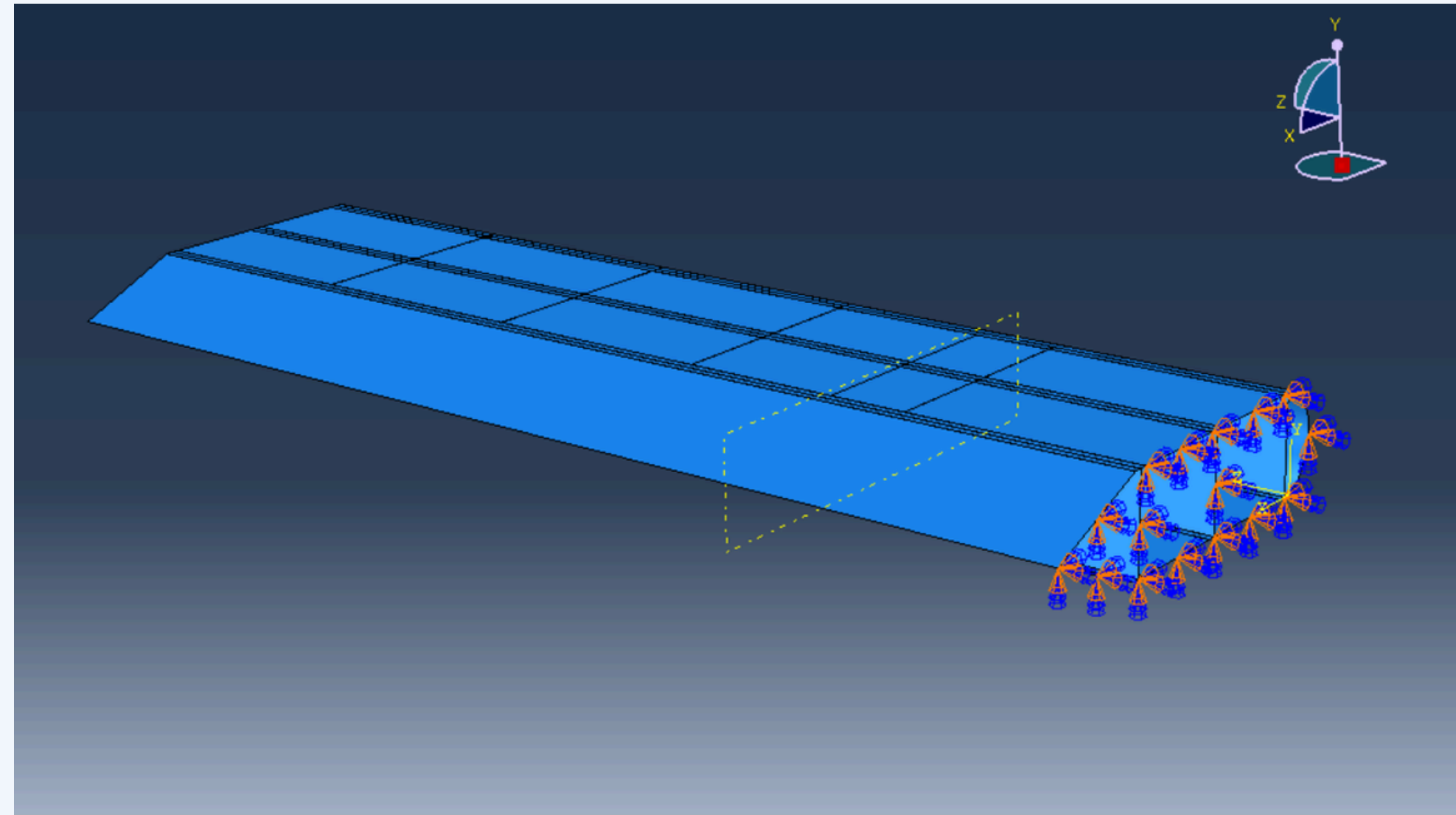
Mesh is sufficiently refined to capture stress variations while maintaining computational efficiency.

Boundary Conditions

Wingbox root is fixed (ENCASTRE condition)

Constraints Applied

- Translational DOF:
 - $U_1 = U_2 = U_3 = 0$
- Rotational DOF:
 - $UR_1 = UR_2 = UR_3 = 0$



All degrees of freedom are constrained at the root, representing rigid attachment to the fuselage.

Loading Conditions

1. Engine Load (Concentrated Force)

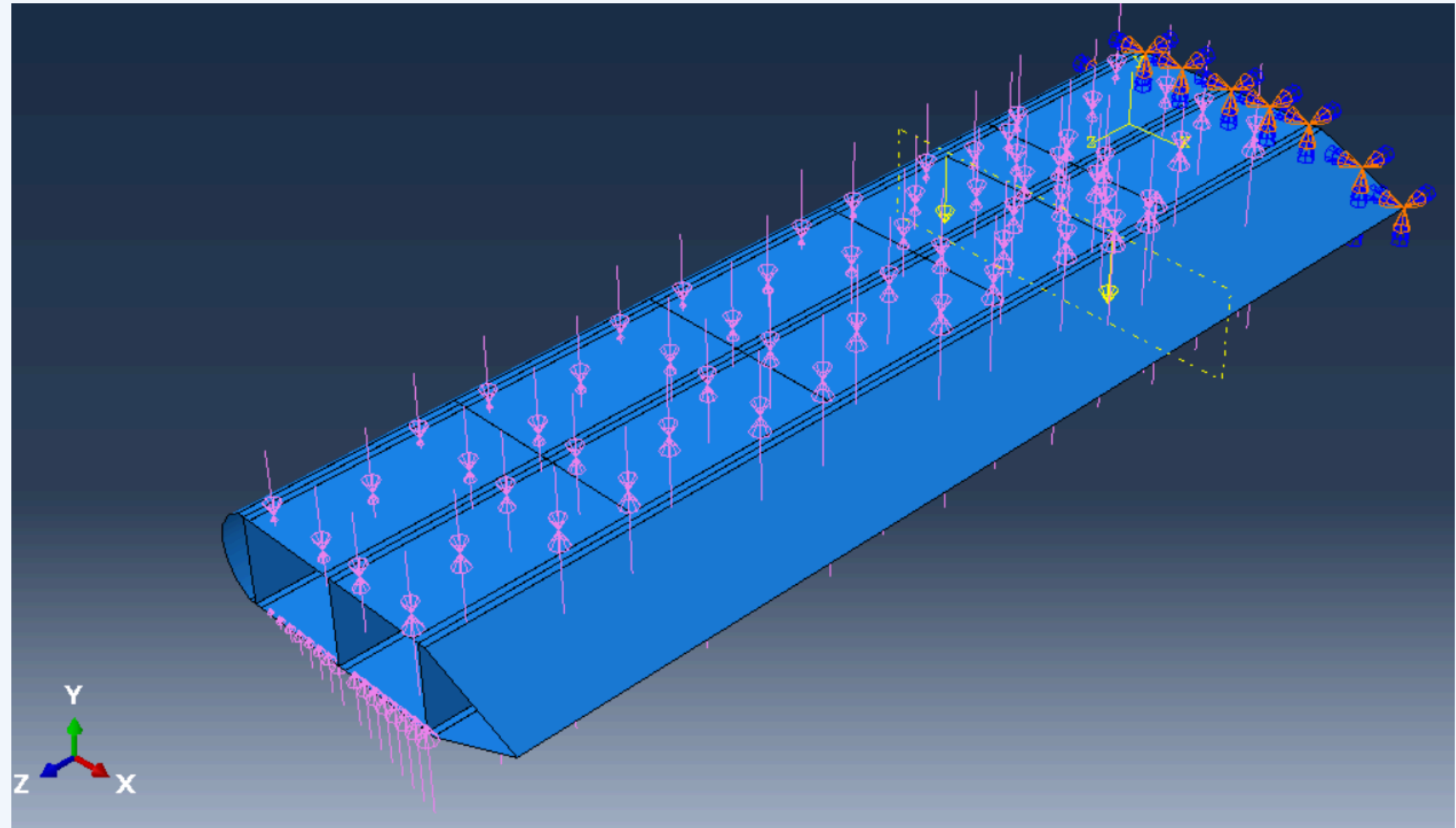
- Applied as point load
- Location: ~4 m from root

2. Pressure Load (Uniform)

- Applied on upper surface
- Distributed load

3. Lift Load (Distributed)

- Applied as linearly varying pressure
- Implemented using expression field
- Represents aerodynamic lift



Analysis Type

Type of Analysis

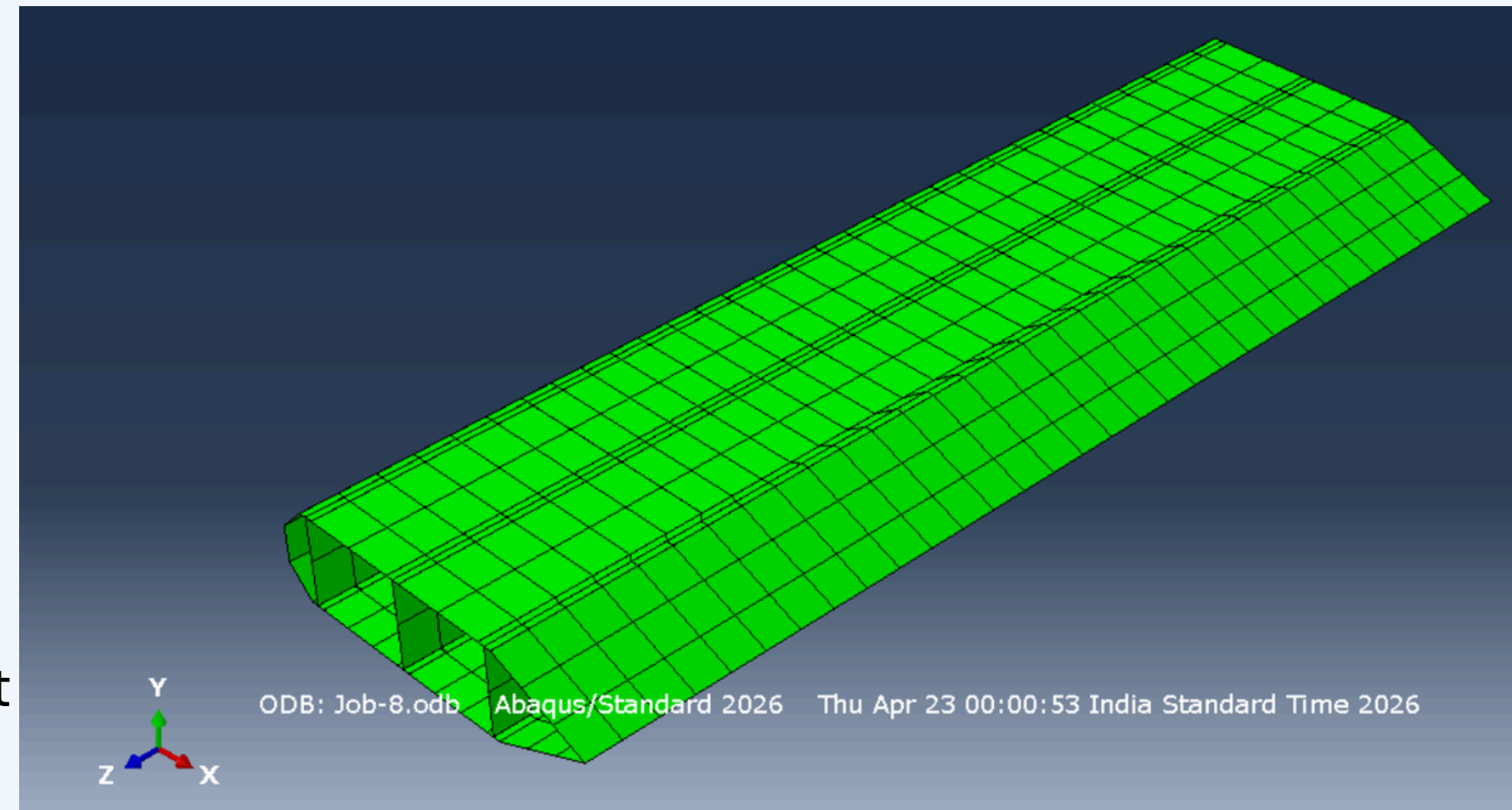
- Static General Analysis (Abaqus/Standard)

Assumptions

- Linear elastic material behavior
- Small deformations
- Loads applied gradually (ramp loading)

Why this analysis?

Suitable for evaluating stress and displacement under steady loading conditions"



Stress Distribution (Von Mises Stress)

Results:

Maximum von Mises stress:

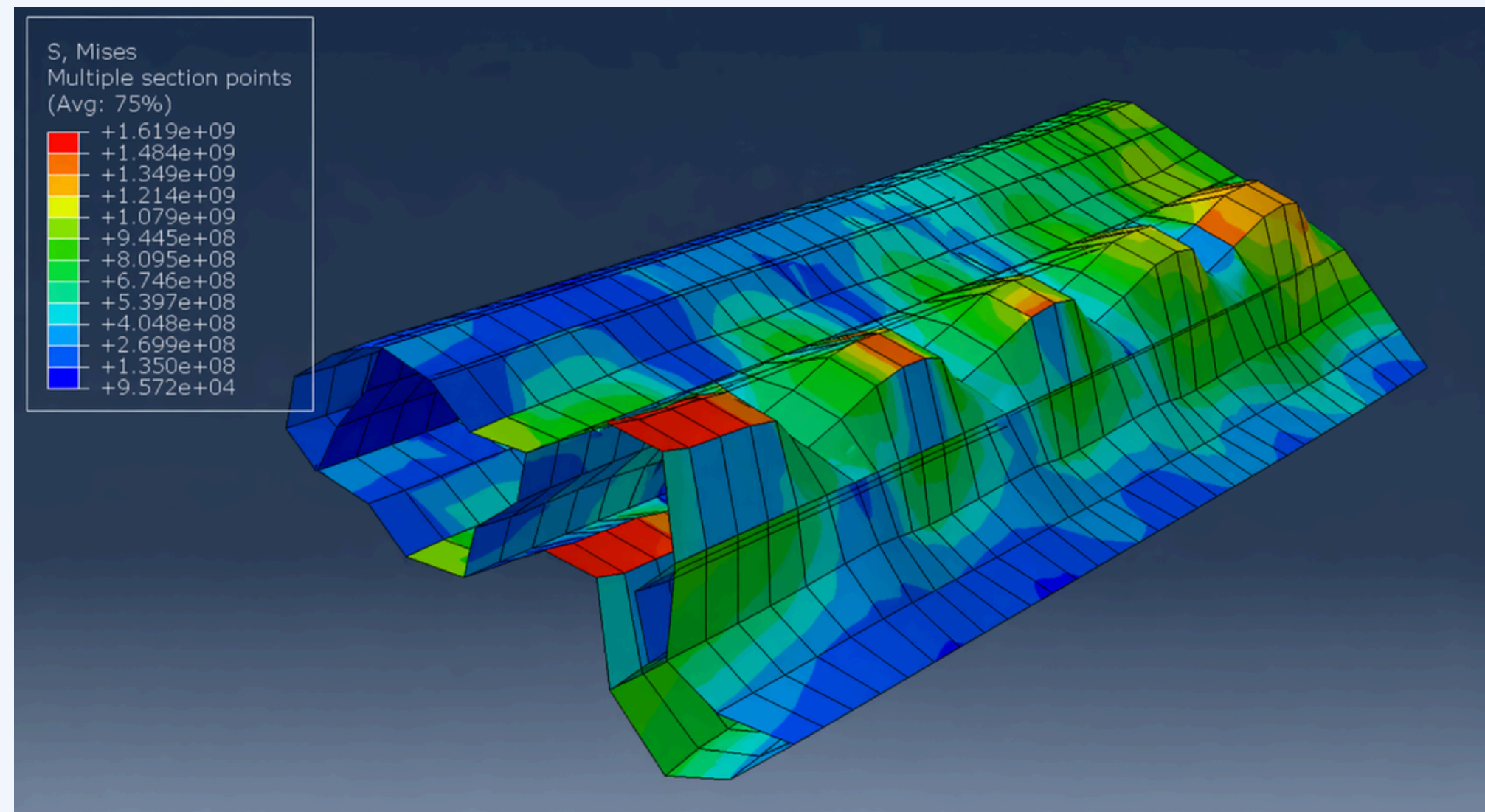
$$\sigma_{max} \approx 2.086 \times 10^8 \text{ Pa} = 208 \text{ MPa}$$

Material Limit (Aluminium)

$$\sigma_y = 310 \text{ MPa}$$

$$\sigma_{max} < \sigma_y \rightarrow \text{SAFE}$$

Structure is safe under applied loading



Maximum stress occurs in the upper skin region, especially between ribs

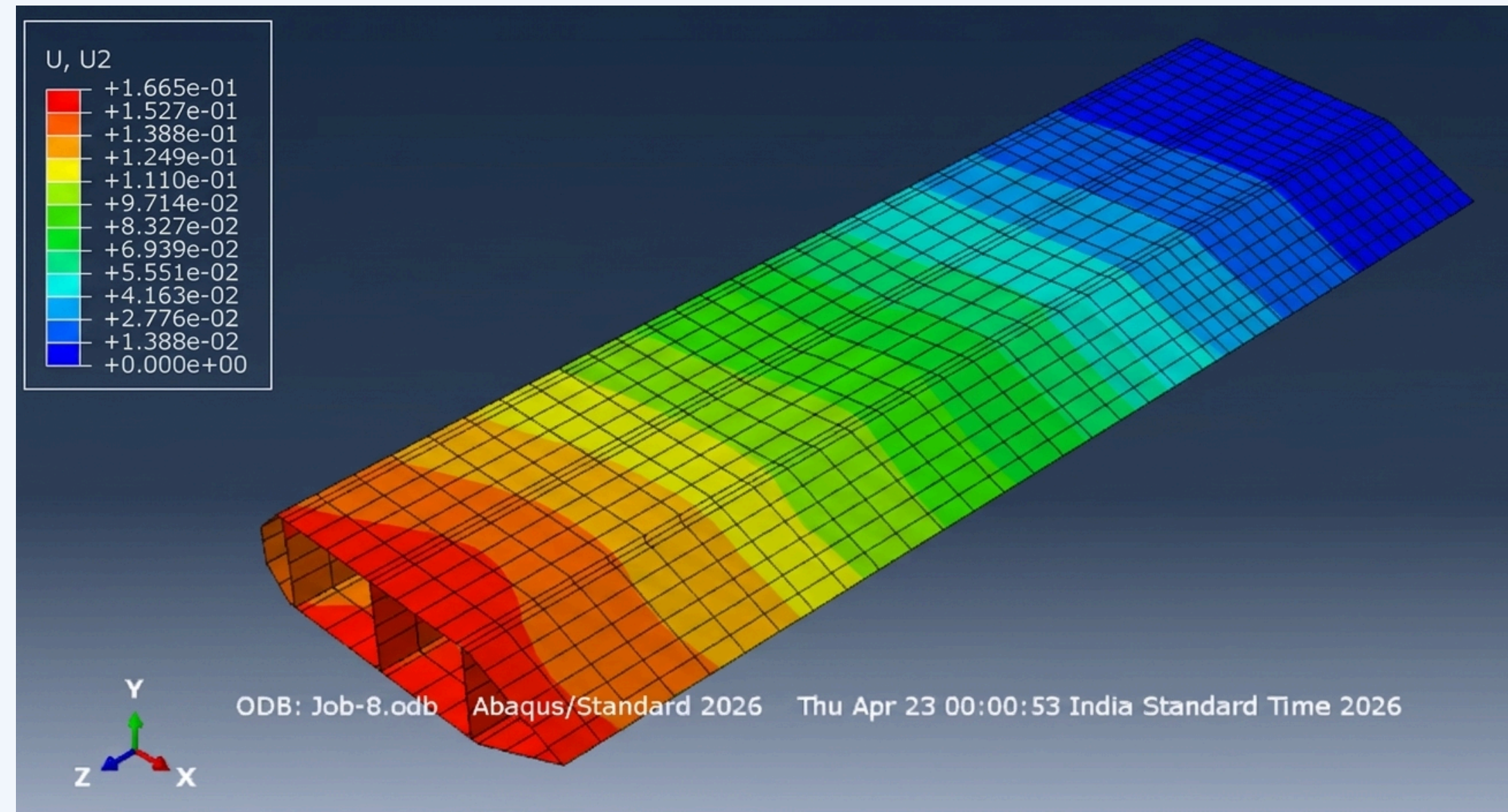
Displacement Results

Maximum displacement (U2) ≈ 6 mm

Maximum displacement occurs at the free end (wing tip) .

Deformation pattern matches expected cantilever beam behavior .

(Note: deformation shown is scaled for visualization)



Reaction Forces and Composite Failure Analysis

REACTION FORCES AT ROOT (RF2)

Extraction Method

- Reaction forces obtained from:
 - ODB Field Output
 - Variable: RF2 (vertical direction)
- Values extracted at:
 - All nodes at fixed root

Total Reaction Force

$$|F| = 3.2365 \times 10^6 \text{ N}$$

COMPOSITE FAILURE ANALYSIS (HASHIN CRITERIA)

Failure Criteria Used

- Hashin Failure Theory
- Evaluates:
 - Fiber tension
 - Fiber compression
 - Matrix tension
 - Matrix compression

Results

- All failure indices: <1
No failure observed

Composite material is safe under applied loading

Mass and Structural Observations

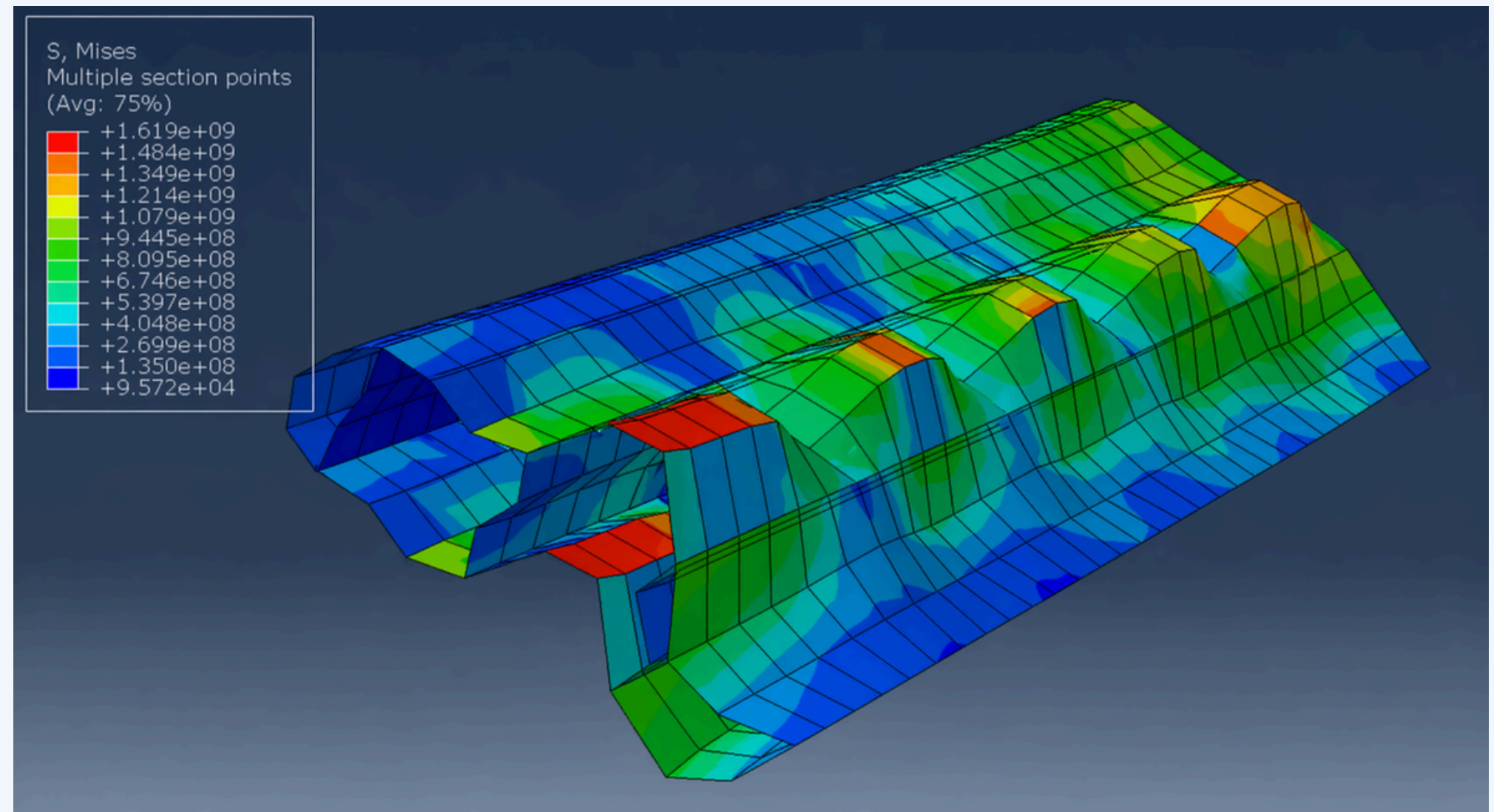
Mass of Structure: $m \approx 5.67 \times 10^4 \text{ kg}$

The structural mass is relatively high and requires optimization

Stress Behavior

- Maximum stress observed:
 - Upper skin
 - Between ribs

Skin panels are critical regions under compressive loading



Skin buckling

Objective:

Determine required skin thickness to prevent buckling

$$\sigma_{cr} \geq \sigma$$

Buckling formula::

$$\sigma_{cr} = \frac{k\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b}\right)^2 \quad \rightarrow \quad t = b \sqrt{\frac{12(1-\nu^2) \sigma}{k\pi^2 E}}$$

Substituting values:

- $b = 0.3 \text{ m}$
- $k = 4$
- $E = 70 \text{ GPa}$
- $\nu = 0.33$
- $\sigma = 208 \text{ MPa}$

$$t_{\min} \approx 3.1 \text{ mm}$$

Adopted thickness = 3.5 mm

$$MS \approx 0.30 \quad (\text{Safe})$$

Spar web buckling

Buckling stress:

$$\sigma_{cr} = \frac{k\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b}\right)^2$$

Assumed Parameters:

- Boundary condition → simply supported
- Buckling coefficient → $k \approx 6$
- Web thickness → $t = 3 \text{ mm}$
- Web height → $b = 0.6 \text{ m}$ (assumed)
- Material → Aluminium
- Buckling stress (initial):
 - **$\sigma_{cr} \approx 110 \text{ MPa}$**
 - Applied stress:
 - **$\sigma \approx 208 \text{ MPa}$**
- Initial condition:
 - $\sigma_{cr} < \sigma \rightarrow$ Not sufficient

Design modification:

Increase web thickness

- **$t = 3 \text{ mm} \rightarrow 4.5 \text{ mm}$**

Updated buckling stress:

$\sigma_{cr} \approx 230\text{--}250 \text{ MPa}$

Comparison:

$\sigma_{cr} > \sigma \rightarrow \text{SAFE}$

- $MS \approx 0.2\text{--}0.3$

Spar flange buckling

Buckling stress:

$$\sigma_{cr} = \frac{k\pi^2 E}{12(1-\nu^2)} \left(\frac{t}{b}\right)^2$$

Given Parameters:

- $k = 4$
- $t = 4 \text{ mm}$
- $b = 0.1 \text{ m}$
- Material: Aluminium

Buckling stress:

$\sigma_{cr} \approx 420 \text{ MPa}$

Applied stress:

$\sigma \approx 208 \text{ MPa}$

Comparison:

$\sigma_{cr} > \sigma \rightarrow \text{SAFE}$

Margin of Safety:

$MS \approx 1.0\text{--}1.2$

Flange is structurally efficient and not critical in buckling

Crippling

Crippling Stress Formula

$$\sigma_{crip} = C \sigma_y \left(\frac{t}{b} \right)^m$$

Given Parameters:

- $\sigma_y = 300 \text{ MPa}$
- $C = 0.9$
- $m = 0.6$
- $t = 4 \text{ mm}$
- $b = 0.1 \text{ m}$

Crippling stress:

$$\sigma_{crip} \approx 260 \text{ MPa}$$

Applied stress:

$$\sigma \approx 208 \text{ MPa}$$

Comparison:

$$\sigma_{crip} > \sigma \rightarrow \text{SAFE}$$

Margin of Safety:

$$MS \approx 0.25-0.30$$

Crippling is not a governing failure mode for the spar flange

Conclusion

- A finite element model of the wingbox was developed in Abaqus
- Stress and displacement results indicate:
 - Structure is safe under static loading
 - Maximum stress occurs in upper skin between ribs
 - Deformation follows expected cantilever behavior
- Buckling analysis shows:
 - Skin and spar web are critical components
 - Design modifications were required
- Design improvements:
 - Skin thickness increased → safe in buckling
 - Spar web thickness increased → safe in buckling
- Spar flange:
 - Safe in both buckling and crippling
 - Not a critical component
- Final design satisfies strength and stability requirements

Usage of AI

Used chatGpt to understand the (a) buckling of skin, (b) buckling of spar web, (c) buckling of spar flange, and (d) crippling of spar flange phenomenon.

Prompt: Explain buckling of skin, spar web and flange and crippling of spar flange for wingbox. Also give formulae.

During the job submission in abaqus , error occurred due to missing assignment of sections which led to aborted the job. So we used ai to analyse the error in job.inp file and also snipping error dialogue box.

Prompt: This is the input file for our abaqus wingbox model. Job is getting aborted. Please make necessary changes. Pls help



Thank You